

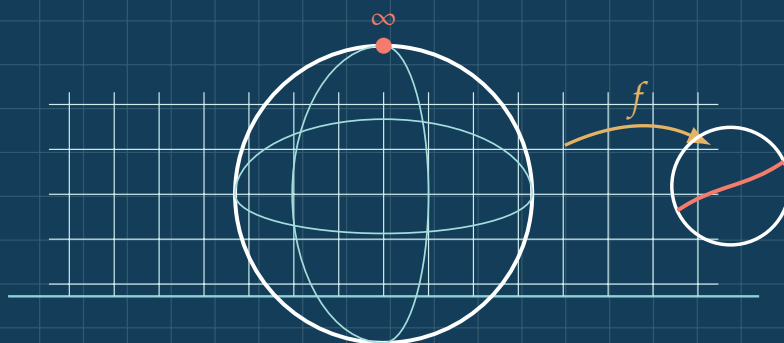
A VISUAL GUIDE TO COMPLEX PROJECTIVE GEOMETRY

Möbius Transformations

and the

Schwarzian Derivative

Circles, cross-ratios, osculating maps,
and the first genuinely projective derivative

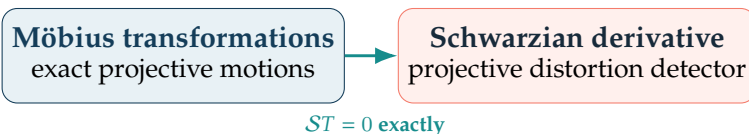


Built from definitions, pictures, proofs, and worked examples

$\hat{\mathbb{C}} \rightarrow \hat{\mathbb{C}}$

The central idea

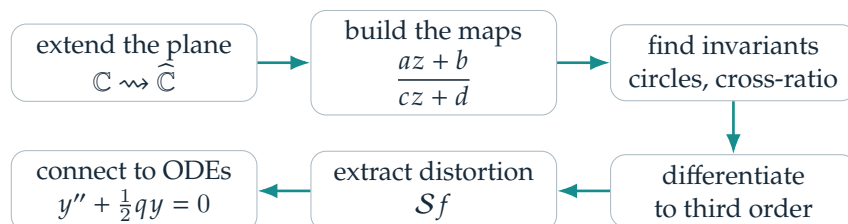
Möbius transformations are the exact symmetries of the Riemann sphere. The Schwarzian derivative measures the part of a holomorphic map that cannot be explained by those symmetries. Put differently:



One sentence to remember

Ordinary derivatives compare a map with an affine approximation; the Schwarzian compares it with its best local Möbius approximation.

Road map



Notation and standing assumptions

$\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$	the extended complex plane, identified with the Riemann sphere
$\mathbb{D}$	the unit disk $\{z : z < 1\}$
$\mathbb{H}$	the upper half-plane $\{z : \text{Im } z > 0\}$
Sf	the Schwarzian derivative of a locally univalent holomorphic map f

"Locally univalent" means $f'(z) \neq 0$. This is the natural setting for the Schwarzian because its formula divides by f' . All geometric statements about lines tacitly regard a line as a circle through ∞ .

A useful hierarchy

Euclidean maps preserve lengths and angles. Similarities preserve ratios of lengths and angles. Möbius maps preserve angles and the cross-ratio. General holomorphic maps preserve angles locally, but usually not the cross-ratio.

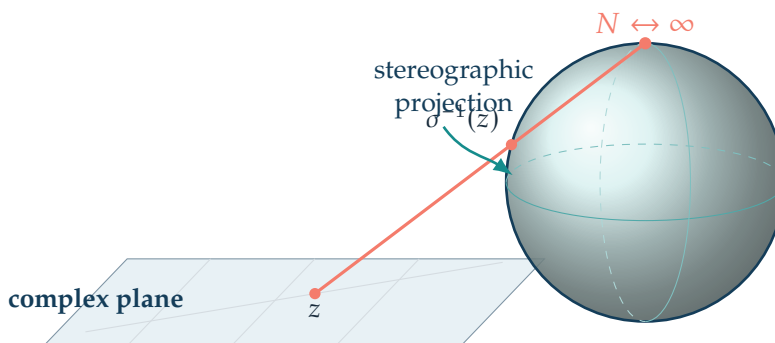
PART I Möbius transformations

Why add one point at infinity?

The function $z \mapsto 1/z$ is almost a symmetry of the complex plane: it merely fails at 0, and it sends very large numbers toward 0. Both issues disappear if we adjoin a single point ∞ and decree

$$\frac{1}{0} = \infty, \quad \frac{1}{\infty} = 0.$$

The resulting space $\widehat{\mathbb{C}}$ is not an artificial patch. It is a sphere.



Stereographic projection from the north pole N gives a one-to-one correspondence between the sphere and $\widehat{\mathbb{C}}$. Neighborhoods of ∞ are the complements of large disks. This makes limits at infinity precise:

$$f(\infty) = \lim_{z \rightarrow \infty} f(z), \quad \lim_{z \rightarrow \infty} f(z) = L \iff \lim_{w \rightarrow 0} f(1/w) = L.$$

Geometric payoff

On the sphere, a Euclidean line closes up by passing through ∞ . Thus circles and lines become a single family: the *generalized circles*.

Conformality

Stereographic projection preserves angles. Complex differentiation also preserves angles wherever $f' \neq 0$: infinitesimally, f acts as multiplication by $f'(z)$, which is a rotation followed by a scaling. Möbius maps are the global, bijective conformal self-maps of $\widehat{\mathbb{C}}$.

The maps and their algebra

Definition

For $a, b, c, d \in \mathbb{C}$ with $ad - bc \neq 0$, a **Möbius transformation** is

$$T(z) = \frac{az + b}{cz + d}.$$

At the exceptional points, $T(-d/c) = \infty$ when $c \neq 0$, and $T(\infty) = a/c$ when $c \neq 0$; if $c = 0$, then $T(\infty) = \infty$.

The determinant condition prevents numerator and denominator from being proportional. Direct differentiation gives

$$T'(z) = \frac{ad - bc}{(cz + d)^2},$$

so T is locally conformal wherever it is finite.

Matrices, but only up to scale

Associate to T the matrix

$$A_T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Multiplying all four entries by the same $\lambda \neq 0$ does not change T . Composition of transformations corresponds exactly to matrix multiplication:

$$A_{T \circ U} = A_T A_U.$$

Thus the Möbius group is $\text{PGL}(2, \mathbb{C})$. Equivalently it is $\text{PSL}(2, \mathbb{C})$, after rescaling a representing matrix to determinant 1.

Inverse without guesswork

If $T(z) = (az + b)/(cz + d)$, invert its matrix:

$$A_T^{-1} \sim \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Therefore

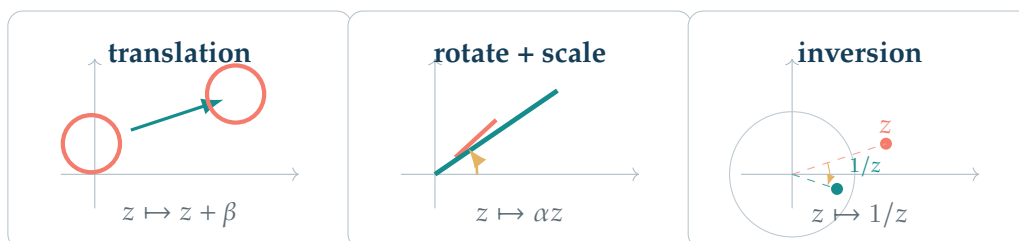
$$T^{-1}(w) = \frac{dw - b}{-cw + a}.$$

Four atomic motions

Every Möbius map can be built from translations, rotations/dilations, and inversion. When $c \neq 0$,

$$\frac{az + b}{cz + d} = \frac{a}{c} - \frac{ad - bc}{c(cz + d)}.$$

Reading from right to left: translate/scale z to $cz + d$, invert, scale, then translate.



The only nonlinear-looking atom is inversion. For $z = re^{i\theta}$, it sends z to $r^{-1}e^{-i\theta}$: invert the radius and reflect the angle. Its geometry is why circles and lines are the right objects to track.

Circles become circles - sometimes through infinity

A generalized circle has an equation

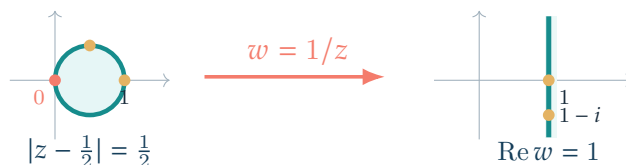
$$A|z|^2 + Bz + \bar{B}\bar{z} + C = 0, \quad A, C \in \mathbb{R},$$

not all coefficients zero. If $A \neq 0$ this is a circle; if $A = 0$ it is a line. Substituting $z = T^{-1}(w)$ and clearing $|-cw + a|^2$ produces another equation of the same form. Therefore:

Generalized-circle theorem

Every Möbius transformation sends generalized circles to generalized circles. A source circle becomes a line precisely when it contains the pole $T^{-1}(\infty)$.

A CIRCLE THROUGH THE POLE OPENS INTO A LINE

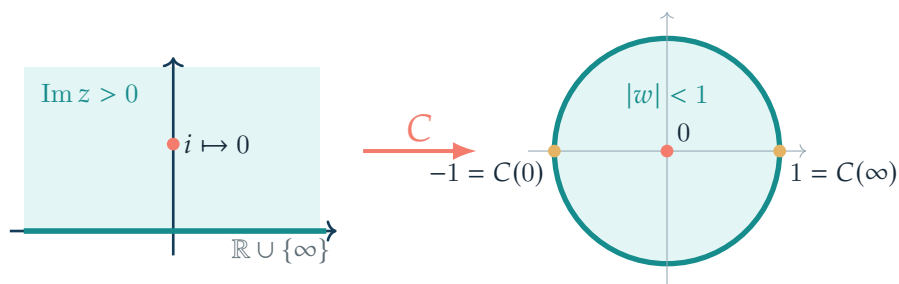


The Cayley transform

The map

$$C(z) = \frac{z - i}{z + i}$$

sends the upper half-plane $\mathbb{H}$ onto the unit disk $\mathbb{D}$.



Why? For real x , $|x - i| = |x + i|$, so $|C(x)| = 1$: the boundary line maps to the boundary circle. The test point i maps to 0, selecting the interior. This “boundary plus one test point” method is often faster than solving an inequality.

Map a general half-plane to the disk

Suppose a half-plane is bounded by a generalized circle L . First choose a Möbius map taking three points of L to $-1, 1, \infty$; then L becomes the real axis. Compose with the Cayley transform, and use one interior test point to choose between $\mathbb{D}$ and its exterior.

Angles survive

Because $T'(z) \neq 0$, two smooth curves meeting at z retain their oriented angle after applying T . Their global shapes may change dramatically - a line may bend into a circle - but their infinitesimal crossing geometry does not.

Algebra confirms the picture

Solving $w = (z - i)/(z + i)$ for z gives

$$z = C^{-1}(w) = i \frac{1 + w}{1 - w}.$$

Multiplying numerator and denominator by $1 - \bar{w}$ shows that

$$\operatorname{Im} C^{-1}(w) = \frac{1 - |w|^2}{|1 - w|^2}.$$

Thus $|w| < 1$ is exactly equivalent to $\operatorname{Im} z > 0$; the boundary $|w| = 1$ corresponds to the real axis together with ∞ .

Three points determine the map

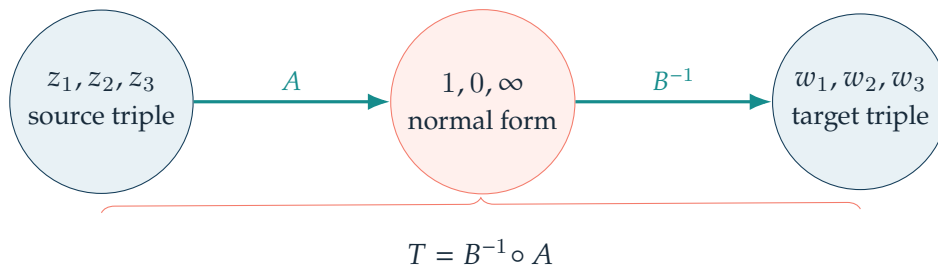
No nonidentity Möbius transformation can fix three distinct points. Conversely, given distinct $z_1, z_2, z_3 \in \widehat{\mathbb{C}}$ and distinct $w_1, w_2, w_3 \in \widehat{\mathbb{C}}$, there is a unique Möbius map with $T(z_j) = w_j$.

The natural computational device is the **cross-ratio**:

$$[z, z_1; z_2, z_3] = \frac{(z - z_2)(z_1 - z_3)}{(z - z_3)(z_1 - z_2)}.$$

This normalization sends (z_1, z_2, z_3) to $(1, 0, \infty)$. Cross-ratios are Möbius invariant:

$$[Tz, Tz_1; Tz_2, Tz_3] = [z, z_1; z_2, z_3].$$



Worked map: $(0, 1, \infty) \mapsto (-1, 0, 1)$

Let $T(z) = (az + b)/(cz + d)$. The conditions $T(0) = -1$ and $T(1) = 0$ give $b = -d$ and $a = -b = d$. The condition $T(\infty) = 1$ gives $a/c = 1$, hence $c = d$. Up to scale,

$$T(z) = \frac{z - 1}{z + 1}.$$

The pole -1 is the source point sent to infinity.

Fixed points and normal forms

Fixed points solve

$$z = \frac{az + b}{cz + d} \iff cz^2 + (d - a)z - b = 0.$$

So a nonidentity Möbius map has one or two fixed points on $\widehat{\mathbb{C}}$ (counting multiplicity). Conjugation reveals the dynamics:

- Two distinct fixed points can be moved to 0 and ∞ , after which the map is $z \mapsto \lambda z$.
- One double fixed point can be moved to ∞ , after which the map is $z \mapsto z + 1$.

Conjugacy is a change of projective coordinates. It does not alter the essential dynamics - only the viewpoint.

Part I in one line

Möbius transformations are fractional-linear, matrix-controlled, conformal symmetries of $\widehat{\mathbb{C}}$; they preserve generalized circles and cross-ratios, and three point images determine them completely.

PART II The Schwarzian derivative

Why ordinary derivatives are not enough

For a locally univalent holomorphic function f ,

$$f(z+h) = f(z) + f'(z)h + \frac{1}{2}f''(z)h^2 + \frac{1}{6}f'''(z)h^3 + O(h^4).$$

The first derivative records local rotation and scaling. The ratio f''/f' , called the pre-Schwarzian, records how that affine behavior changes. But even a Möbius transformation generally has $T''/T' \neq 0$, so this still mistakes projective motion for genuine distortion.

We need a third-order expression that vanishes for *every* Möbius map.

Definition

The **Schwarzian derivative** of a locally univalent holomorphic map f is

$$\mathcal{S}f = \frac{f'''}{f'} - \frac{3}{2} \left(\frac{f''}{f'} \right)^2$$

or, equivalently,

$$\mathcal{S}f = \left(\frac{f''}{f'} \right)' - \frac{1}{2} \left(\frac{f''}{f'} \right)^2.$$

Why the coefficient is exactly 3/2

For $T(z) = (az + b)/(cz + d)$,

$$\frac{T''}{T'} = -\frac{2c}{cz + d}, \quad \frac{T'''}{T'} = \frac{6c^2}{(cz + d)^2}.$$

Therefore

$$\mathcal{S}T = \frac{6c^2}{(cz + d)^2} - \frac{3}{2} \frac{4c^2}{(cz + d)^2} = 0.$$

The cancellation is not cosmetic: 3/2 is forced by projective invariance.

Vanishing theorem

On a connected domain, $\mathcal{S}f = 0$ if and only if f is the restriction of a Möbius transformation.

Indeed, setting $u = f''/f'$ turns $\mathcal{S}f = 0$ into $u' = u^2/2$. Its solutions are $u = 0$ or $u = -2/(z - z_0)$; integrating twice yields affine or fractional-linear maps.

Dimensional clue

Both terms in $\mathcal{S}f$ have units of “inverse length squared.” This is why the Schwarzian naturally behaves like a quadratic differential $q(z) dz^2$, not like an ordinary scalar function.

The best local Möbius approximation

Fix z_0 with $f'(z_0) \neq 0$. There is a unique Möbius map M whose value, first derivative, and second derivative agree with those of f at z_0 . Writing $h = z - z_0$, it is

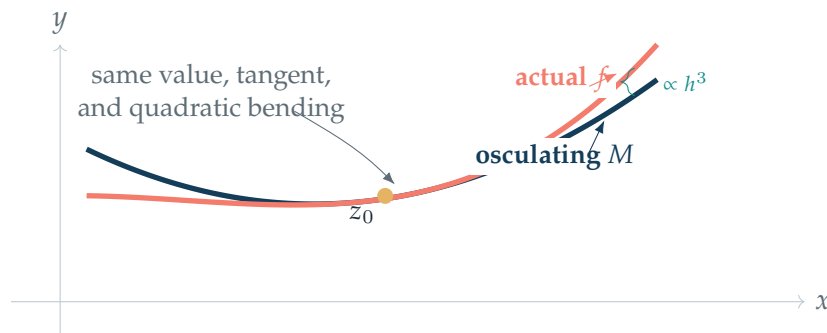
$$M(z) = f(z_0) + \frac{f'(z_0)h}{1 - \frac{f''(z_0)}{2f'(z_0)}h}.$$

Its Taylor series matches f through order two. The first mismatch is

The geometric meaning of Sf

$$f(z) - M(z) = \frac{f'(z_0)}{6} Sf(z_0)(z - z_0)^3 + O((z - z_0)^4).$$

The Schwarzian is the normalized cubic error after subtracting the best local projective motion.



A hierarchy of local models

constant	matches $f(z_0)$	error begins at order 1
affine	matches $f(z_0), f'(z_0)$	error begins at order 2
Möbius	matches $f(z_0), f'(z_0), f''(z_0)$	error begins at order 3

This is the conceptual upgrade: the Schwarzian does for projective geometry what f'' does for affine geometry.

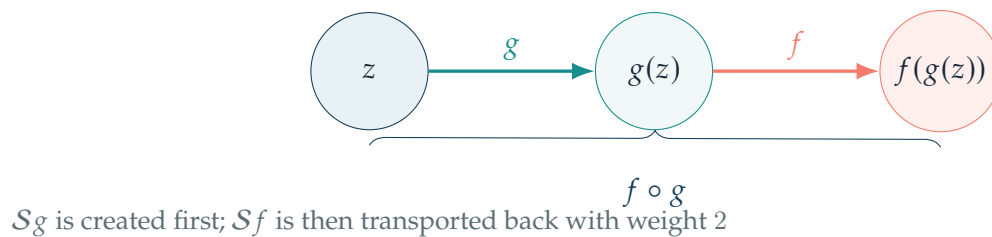
The chain rule and projective invariance

The decisive identity is

Schwarzian chain rule

$$\mathcal{S}(f \circ g) = ((\mathcal{S}f) \circ g)(g')^2 + \mathcal{S}g.$$

The factor $(g')^2$ is the transformation law of a quadratic differential. The two terms also have a clean interpretation: distortion contributed by f , pulled back through g , plus distortion contributed by g itself.



Two consequences should become automatic:

1. Postcomposition invariance. If T is Möbius, then

$$\mathcal{S}(T \circ f) = \mathcal{S}f.$$

Changing projective coordinates in the target does not change the measured distortion.

2. Precomposition covariance. If T is Möbius, then

$$\mathcal{S}(f \circ T) = ((\mathcal{S}f) \circ T)(T')^2.$$

Changing coordinates in the source transforms the Schwarzian as a quadratic differential.

One computation becomes a family

Because $\mathcal{S}(e^z) = -\frac{1}{2}$, postcomposing e^z with any Möbius map leaves its Schwarzian unchanged. For example,

$$F(z) = \frac{e^z - 1}{e^z + 1} \implies \mathcal{S}F = -\frac{1}{2}.$$

There is no need to differentiate F three times.

Inverse rule

Apply the chain rule to $f \circ f^{-1} = \text{id}$:

$$\mathcal{S}(f^{-1})(w) = -\mathcal{S}f(f^{-1}(w))((f^{-1})'(w))^2.$$

The negative sign says that undoing a map undoes its projective distortion.

The hidden linear differential equation

The Schwarzian looks nonlinear, but it is the projectivization of a linear second-order differential equation. Let q be holomorphic and solve

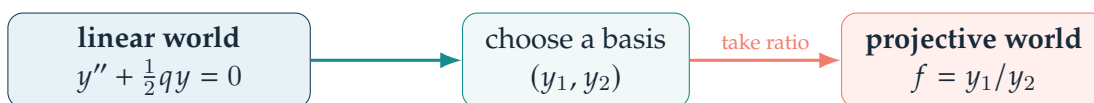
$$y'' + \frac{1}{2}q(z)y = 0.$$

If y_1, y_2 are linearly independent solutions, then

ODE-Schwarzian correspondence

$$f = \frac{y_1}{y_2} \implies Sf = q.$$

Conversely, every locally univalent solution of $Sf = q$ is locally such a ratio.



basis change $\Rightarrow$ Möbius postcomposition

Here is the calculation. The Wronskian $W = y_1' y_2 - y_1 y_2'$ is constant, so

$$f' = \frac{W}{y_2^2}, \quad \frac{f''}{f'} = -2 \frac{y_2''}{y_2}.$$

Substitution into the Schwarzian gives

$$Sf = -2 \frac{y_2''}{y_2} = q.$$

Changing the solution basis by $\begin{pmatrix} \tilde{y}_1 \\ \tilde{y}_2 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$ changes the ratio to

$$\tilde{f} = \frac{af + b}{cf + d}.$$

That is precisely Möbius postcomposition, which explains why Sf is independent of the chosen basis.

Constant Schwarzian

For $q = 2k^2$, the ODE is $y'' + k^2 y = 0$. Taking $y_1 = \sin(kz)$ and $y_2 = \cos(kz)$ gives

$$f(z) = \tan(kz), \quad Sf = 2k^2.$$

Every other solution of $Sf = 2k^2$ is a Möbius transform of $\tan(kz)$.

This bridge is the source of the Schwarzian's importance in conformal mapping, projective structures, monodromy, Teichmüller theory, and one-dimensional dynamics.

A gallery of Schwarzians

The quickest calculations use the logarithmic derivative $P_f = f''/f'$ and

$$\mathcal{S}f = P'_f - \frac{1}{2}P_f^2.$$

map $f(z)$	$\mathcal{S}f(z)$	geometric reading
$\frac{az+b}{cz+d}$	0	pure projective motion
z^n	$\frac{1-n^2}{2z^2}$	branching produces an inverse-square signal
e^{az}	$-\frac{a^2}{2}$	constant projective distortion
$\tan(az)$	$2a^2$	constant distortion with opposite sign
$\log z$	$\frac{1}{2z^2}$	inverse-square distortion near 0

Worked example: the power map

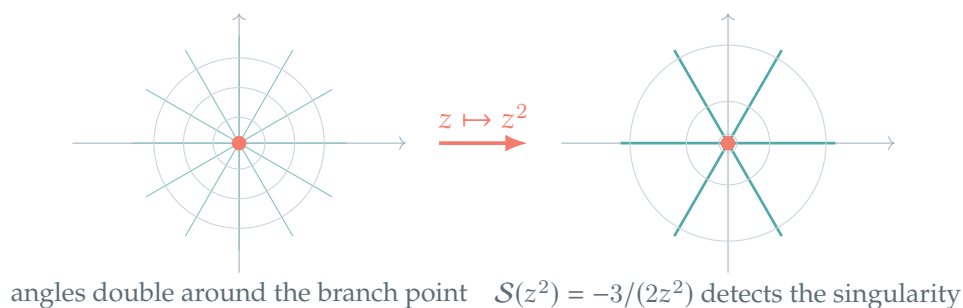
For $f(z) = z^n$,

$$\frac{f''}{f'} = \frac{n-1}{z},$$

so

$$\mathcal{S}(z^n) = -\frac{n-1}{z^2} - \frac{1}{2} \frac{(n-1)^2}{z^2} = \frac{1-n^2}{2z^2}.$$

For $n = 1$ the Schwarzian vanishes, as it must. For $n \neq 1$, the pole at the branch point records the failure of local univalence there.



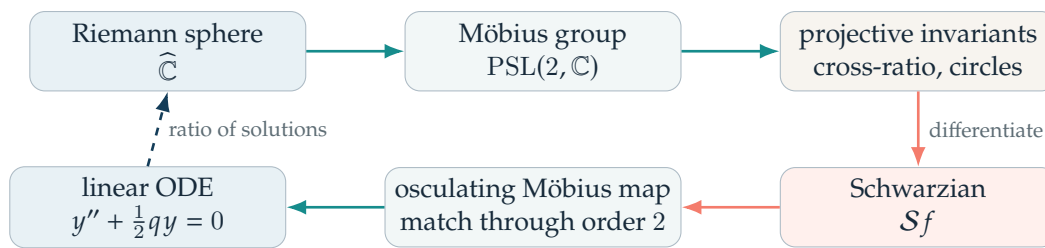
A global consequence: univalence

The Schwarzian controls how far a map can drift from projective behavior. One sharp landmark is **Ne-hari's criterion**: if f is locally univalent on $\mathbb{D}$ and

$$|\mathcal{S}f(z)|(1-|z|^2)^2 \leq 2 \quad (z \in \mathbb{D}),$$

then f is univalent on $\mathbb{D}$. The hyperbolic factor $(1-|z|^2)^2$ makes the bound invariant under automorphisms of the disk.

How the two subjects lock together



The synthesis

Möbius transformations provide the coordinate changes of one-dimensional complex projective geometry. The Schwarzian is the differential invariant that survives those changes. Its vanishing recognizes projective maps; its chain rule describes coordinate change; and its ODE realization linearizes the equation $Sf = q$.

A practical workflow

When facing a concrete problem, ask these questions in order:

1. **Normalize three points.** Can a Möbius map move the important points to 0, 1, ∞ , or move the domain to $\mathbb{D}$ or $\mathbb{H}$?
2. **Track generalized circles.** Where is the pole, and which side of a boundary maps to which side?
3. **Use invariance before differentiating.** Is the function a composition with a known Möbius map?
4. **If $Sf = q$, change viewpoint.** Solve $y'' + \frac{1}{2}qy = 0$ and take a ratio.
5. **Interpret, do not merely compute.** A pole of Sf often marks a critical point, branch point, or singular projective structure.

Common pitfalls

- Forgetting the point ∞ when solving image problems.
- Calling lines exceptional instead of circles through ∞ .
- Treating the matrix of a Möbius map as unique rather than projective.
- Assuming f''/f' vanishes for Möbius maps.
- Omitting $(g')^2$ in the Schwarzian chain rule.
- Using Sf at a critical point where $f' = 0$ without allowing a pole.

Final mental picture

On a tiny neighborhood, replace f by the unique Möbius map with the same value, velocity, and acceleration. Zoom in on their cubic separation. After normalizing by f' , what remains is $Sf/6$.

Compact reference sheet

Möbius toolkit

$$T(z) = \frac{az + b}{cz + d}, \quad ad - bc \neq 0$$

$$T'(z) = \frac{ad - bc}{(cz + d)^2}$$

$$T^{-1}(w) = \frac{dw - b}{-cw + a}$$

Preserves: angles, generalized circles, cross-ratios.

Determined by: images of three distinct points.

Matrix law: composition $\leftrightarrow$ multiplication.

Canonical maps

$$\mathbb{H} \rightarrow \mathbb{D} : \quad z \mapsto \frac{z - i}{z + i}$$

$$(z_1, z_2, z_3) \rightarrow (1, 0, \infty) : \quad z \mapsto [z, z_1; z_2, z_3]$$

Schwarzian toolkit

$$\mathcal{S}f = \frac{f'''}{f'} - \frac{3}{2} \left(\frac{f''}{f'} \right)^2$$

$$\mathcal{S}(f \circ g) = ((\mathcal{S}f) \circ g)(g')^2 + \mathcal{S}g$$

$$\mathcal{S}(T \circ f) = \mathcal{S}f \quad (T \text{ Möbius})$$

$$\mathcal{S}f = 0 \iff f \text{ is Möbius}$$

$$\mathcal{S}(y_1/y_2) = q \quad \text{if} \quad y'' + \frac{1}{2}qy = 0$$

Three short exercises

1. Find the Möbius map taking $(1, i, -1)$ to $(0, 1, \infty)$.
2. Without three rounds of differentiation, find $\mathcal{S}((e^z - 2)/(e^z + 3))$.
3. Solve locally $\mathcal{S}f = 2$ and describe the freedom in the answer.

Hints and answers

1. Normalize with the cross-ratio: $T(z) = \frac{(z-1)(i+1)}{(z+1)(i-1)}$. It sends $1 \mapsto 0$, $i \mapsto 1$, and $-1 \mapsto \infty$.
2. The outer map $w \mapsto (w-2)/(w+3)$ is Möbius, so the answer is $\mathcal{S}(e^z) = -1/2$.
3. Since $y'' + y = 0$, take $y_1 = \sin z$, $y_2 = \cos z$. Thus $f = T \circ \tan z$ for an arbitrary Möbius transformation T .

Further reading

L. V. Ahlfors, *Complex Analysis*; T. Needham, *Visual Complex Analysis*; Z. Nehari, *Conformal Mapping*; and D. Dumas, “Complex Projective Structures,” in the *Handbook of Teichmüller Theory*.

The sphere supplies the symmetries. The Schwarzian reveals what remains.